

PDF Reader Architecture Research

Task ID: ALO-010

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Objective

The objective of this research is to identify the best architecture and Flutter plugins for implementing a smooth and scalable PDF Reader module in the Alogyan Prep application. The solution should support offline reading, bookmarks, progress tracking, continue reading, caching, and a book-like reading experience.

1. PDF Viewing

Plugin: PDFX / Syncfusion PDF Viewer

Purpose:

- Display PDF files inside the application.
- Support smooth scrolling and page navigation.

Architecture

PDF URL / Local PDF



PDFX



Render PDF Pages



User Reads PDF

Recommendation:

- PDFX for lightweight and fast rendering.
 - Syncfusion for advanced features like search and annotations.
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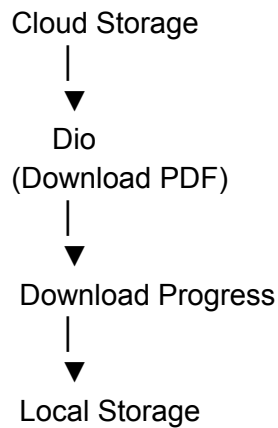
2. PDF Downloading

Plugin: Dio

Purpose:

- Download PDFs from cloud storage.
- Track download progress.

Architecture



Benefits:

- Reliable downloads
 - Progress tracking
 - Better network handling
-

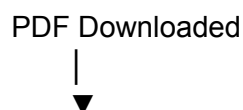
3. Local Storage

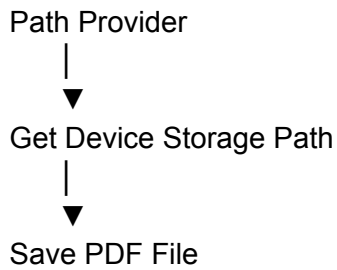
Plugin: Path Provider

Purpose:

- Store downloaded PDFs on the device.
- Support offline reading.

Architecture





Benefits:

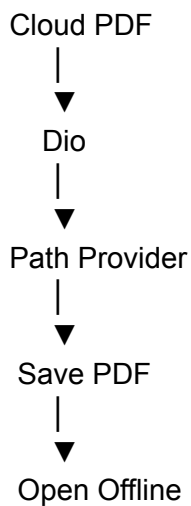
- Organized file management
 - Offline support
-

4. Offline Reading

Purpose:

- Allow users to read PDFs without internet connection.

Architecture



Benefits:

- Works without internet
 - Better user experience
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5. Bookmarks

Plugin: Hive

Purpose:

- Save important pages.
- Quick navigation to saved pages.

Architecture

User Bookmark Page



Save in Hive



Bookmark List



Jump To Page

Example:

- Bookmark Page 25
 - Bookmark Page 80
-

6. Continue Reading

Plugin: Hive

Purpose:

- Resume reading from the last viewed page.

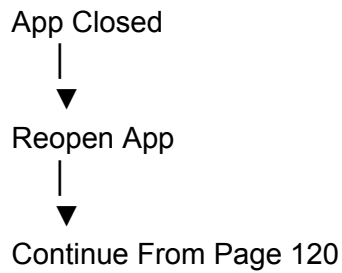
Architecture

User Page 120



Save in Hive





Benefits:

- Better reading continuity
 - Improved user engagement
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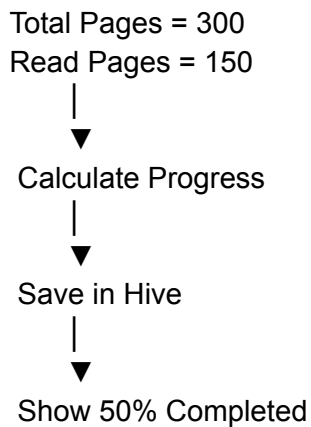
7. Progress Tracking

Plugin: Hive

Purpose:

- Track user reading progress.

Architecture



Benefits:

- Reading analytics
 - Completion tracking
-

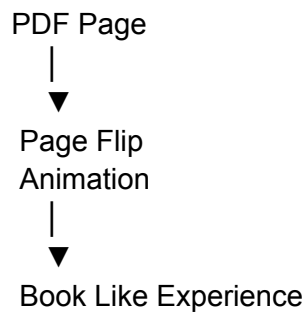
8. Book-like Reading Experience

Plugin: Page Flip

Purpose:

- Provide an Apple Books style reading experience.

Architecture



Benefits:

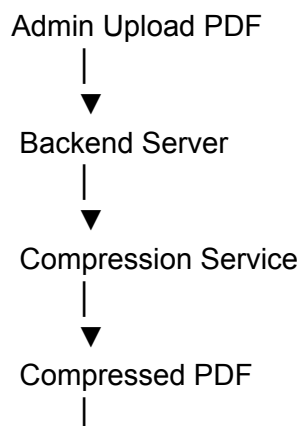
- Smooth page transitions
 - Premium user experience
-

9. PDF Compression

Purpose:

- Reduce PDF size before delivery.

Architecture





Cloud Storage

Benefits:

- Faster downloads
 - Reduced storage cost
 - Better performance
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10. PDF Split & Merge

Solution: Syncfusion PDF Library

Purpose:

- Split large PDFs into chapters.
- Merge multiple PDFs into a single bundle.

Architecture

Chapter 1 PDF

Chapter 2 PDF

Chapter 3 PDF



Merge Service



Master PDF

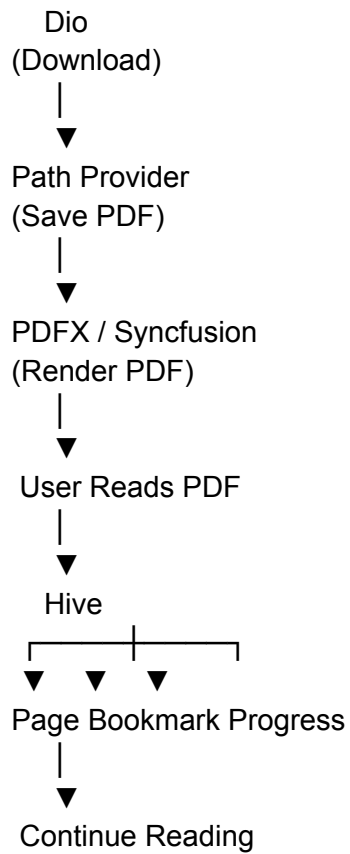
Benefits:

- Better content management
 - Flexible study material creation
-

11. Complete System Architecture

Cloud PDF





Recommended Technology Stack

Feature	Plugin
PDF Rendering	PDFX / Syncfusion
PDF Download	Dio
Local Storage	Path Provider
Bookmarks	Hive
Progress Tracking	Hive
Continue Reading	Hive
Page Animation	Page Flip
PDF Split/Merge	Syncfusion PDF

Conclusion

The proposed architecture uses PDFX/Syncfusion for PDF rendering, Dio for downloading files, Path Provider for local storage, Hive for bookmarks and reading progress, and Page Flip for a premium reading experience. This solution supports offline reading, continue reading, progress tracking, PDF optimization, and future scalability for the Alogyan Prep application.